

Conditions for Social Trajectory Collapse: Agent-Based Simulation of Time-Geographic Trajectory Distributions

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Abstract. Better transportation, communication, and information access are commonly expected to diversify activity-space trajectories, yet they may also concentrate their distribution. We study this concentration with an agent-based model of regions that differ in opportunity, cost, support, and platform visibility. Agents repeatedly reorient across regions, and their choices feed back into regional cost and visibility. Anchored to five urban systems, the model reproduces 2026 regional population shares from 2010 data with small mean absolute errors. The resulting simulations project continued concentration in single-core and radial systems, with substantially weaker concentration in polycentric and federal systems. Concentration and welfare diverge, because an over-expensive dominant region can disperse agents without improving their lives. These findings suggest that social trajectory collapse depends on the interaction between regional opportunity, cost, support, visibility, and urban structure, rather than on connectivity alone.

Keywords: Computational social science · Agent-based modeling · Accessibility · Time-geographic trajectories · Urban concentration

1 Introduction

Better mobility, connectivity, and information access are commonly read as forces for diversity, yet they may instead draw a population’s recurring activity-space trajectories toward a single dominant pattern. What distinguishes one trajectory from another is the spatiotemporal organization of recurring activity, namely the region, opportunity structure, and institutions around which everyday movement is organized. The object of analysis is therefore the population-level distribution of recurring time-geographic trajectories [18, 8], and because individual movement is spatially and temporally regular [14, 29], this distribution is stable enough to ask whether it stays broad or concentrates.

When opportunity and visibility concentrate in a few dominant centers, better connectivity draws agents toward the same center, so whether access diversifies or concentrates depends on how opportunity, cost, and support are arranged

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around it and on the configuration of regions through which movement runs. Existing work supplies the pieces of this arrangement but not their interaction: accessibility research treats access as multidimensional rather than travel speed alone [12, 5], capability and migration research separates a visible opportunity from a reachable one [28, 25, 17], and platform-mediated visibility reweights which opportunities look reachable [6]. Which arrangement concentrates a distribution remains open.

We address this gap with an agent-based simulation model in which regions differ in opportunity, cost, support, visibility, and access, and heterogeneous agents repeatedly reorient and feed back into regional cost and visibility. The model is mechanism-oriented rather than a calibrated forecast; outcomes are summarized through top neighborhood share, tail mass, effective neighborhood count, cost burden, and welfare, so concentration emerges from the dynamics and is not a modeling input.

Connectivity alone barely concentrates trajectories. Its effect reverses with the visibility regime and with region structure, and varying only the regional hierarchy and topology shifts concentration substantially, so single-core and radial systems concentrate where polycentric and federal ones stay distributed. The model reproduces present-day regional population shares within a small error and projects continued concentration in single-core and radial systems. Concentration and welfare diverge, because an over-expensive dominant region can disperse agents without improving their lives.

We make four contributions.

- We reframe social trajectory collapse as concentration of a population over a continuous trajectory index, measured after simulation as an emergent outcome and not imposed as an input category.
- We show that the effect of connectivity is contingent, reversing with the visibility regime and with region structure, so connectivity alone does not account for concentration.
- We separate region configuration from the mechanism settings and show that hierarchy and topology shift concentration on their own.
- We anchor the model to five real urban systems, reproduce present-day regional population shares, and distinguish concentration from welfare.

2 Related Work

2.1 Computational Social Science and Agent-Based Explanation

This work follows computational social science, where traces, networks, experiments, and simulation study processes that resist analytical or observational isolation [21]. Agent-based modeling suits micro-to-macro emergence: under Epstein’s generative standard, a macro pattern is explained when explicit micro rules generate it [9], as Schelling’s segregation model shows for weak local preferences producing strong aggregate segregation [27]. We retain this approach but

apply it to the viability of low-frequency trajectory neighborhoods in a continuous opportunity, cost, and support field, with entities and submodels separated following the ODD protocol [16].

2.2 Social Influence, Cascades, and Platform Amplification

Concentration depends on exposure and recursive influence. Diffusion and threshold models explain how communication channels and local adoption rules produce cascades [15], while complex contagion shows that many behaviors require reinforcement from multiple contacts [3]. Platform settings add visibility feedback: social influence increases inequality and unpredictability in cultural markets [26], and algorithmic feedback can homogenize exposure [4]. Because spatial choice concerns place-bearing opportunities rather than items alone, we model visibility as a region-indexed public salience signal that reweights perceived opportunity, consistent with work on migration aspirations and social-media-shaped opportunity structures [17, 6].

2.3 Spatial Opportunity, Cost Capitalization, and Reproduction

Opportunity, cost, and support fields draw on the capability approach [28] and spatial-opportunity research [11]. Urban economics supplies the cost feedback: amenities capitalize into rents [24], increasing returns support concentration [20], housing supply conditions growth [13], and transport gains are absorbed into housing, service, and status costs [23]. Reproduction is motivated by cultural transmission and institutional persistence [2], and policy enters as a support field that can sustain multiple centers in polycentric systems [22]. We combine these mechanisms in a coupled model evaluated through a common trajectory measure.

3 Method

3.1 Model Overview

We operationalize social trajectory collapse as concentration of a population over recurring time-geographic trajectories in continuous trajectory-index space, following an ODD-style separation of entities, state variables, and submodels [16]. Each agent carries an attribute vector x_i (age, region, income, wealth, attachment, family state, and sensitivities to opportunity, cost, support, connectivity, amplification, and policy); each region carries opportunity, cost, support, density, visibility, housing-supply elasticity, transport access, and reproduction capacity.

Each region holds a field $X_r(t) = (O_r, C_r, S_r, V_r)(t)$ with scalar connectivity κ entering through exposure. Agents perceive it through limited reach, $\tilde{X}_{i,r}(t) = \mathcal{E}_i(X_r(t), \text{access, network, platform visibility})$, where V_r weights exposure and heterogeneity enters through $\mathcal{E}_i$ and the sensitivities. Each agent forms

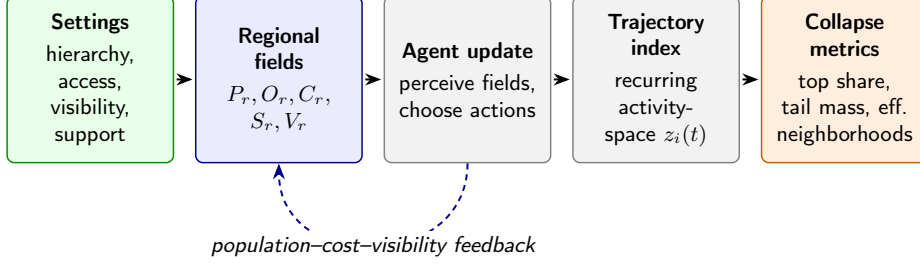


Fig. 1. Model overview. Generator and real-world settings construct regional opportunity–cost–support–visibility fields. Heterogeneous agents perceive those fields through transport access, network reports, and platform visibility, update recurring time-geographic trajectories, and feed population demand back into regional cost and visibility. Collapse is measured after simulation as high top trajectory-neighborhood share, low tail mass, and low effective trajectory-neighborhood count.

a capability index $K_{i,r}(t) = g_i(\tilde{O}, \tilde{C}, \tilde{S}, x_i)(t)$ and chooses by random utility,

$$P(a_i(t) = a) = \frac{\exp(U_{i,a}(t)/\tau)}{\sum_{a'} \exp(U_{i,a'}(t)/\tau)}, \quad (1)$$

over staying, migrating, hybrid access, local investment, and, under reproduction, orienting a child’s trajectory.

Each action update assigns agent i a continuous trajectory index $z_i(t)$ combining its region of recurring activity with its perceived opportunity–cost–support orientation. Partitioning this one-dimensional space into cells, with π_c the share of agents in cell c (an occupied cell being a trajectory neighborhood), we summarize the distribution by

$$q_{\text{top}} = \max_c \pi_c, \quad m_{\text{tail}} = 1 - q_{\text{top}}, \quad N_{\text{eff}} = \exp\left(-\sum_c \pi_c \log \pi_c\right), \quad (2)$$

the largest neighborhood share, the mass outside it, and the entropy-based effective number of occupied neighborhoods. High q_{top} with low m_{tail} and low N_{eff} indicates concentration into a narrow part of trajectory-index space. Cost capitalization closes the loop: as agents crowd a high-opportunity field, its housing, service, schooling, and status costs rise and feed back into the next round of choice, and under reproduction the loop extends through fertility, parent–child transmission, peer density, schools, and media exposure.

3.2 Region Generator

Each region carries a hierarchy rank $r = 1, \dots, R$ and score $h_r \in [0, 1]$, where rank 1 is the primary core and rank R the most peripheral. Population follows a linear control family or, for unequal systems, a Zipf family $P_r = \text{nz}(r^{-\zeta})$ [10] or primate-city family $P_r = \text{nz}(r^{-\zeta} + \pi \mathbb{I}[r=1])$ [19], with $\text{nz}(\cdot)$ normalization.

Opportunity, cost, and visibility each take one of three forms on a field-specific driver set d_r ,

$$f_{\text{lin}}(d_r) = \text{nz}(w_0 + w^\top d_r), \quad (3)$$

$$f_{\text{pow}}(d_r) = \text{nz}\left(\prod_k d_{r,k}^{b_k} / \text{supply}_r^\epsilon\right), \quad (4)$$

$$f_{\text{hinge}}(d_r) = \text{nz}(c_r + \lambda \max(0, u^\top d_r - s_r)^\gamma), \quad (5)$$

with $d_r = (P_r, h_r, \text{hub}_r)$ for opportunity, (P_r, O_r) for cost, and $(O_r, P_r, \text{media}_r)$ for visibility, where supply_r and s_r apply to cost and the hinge form. The linear form scales opportunity with population and hierarchy, the power form scales it superlinearly and capitalizes cost against supply [1, 7, 24, 13], and the hinge form makes visibility threshold-amplified salience [26, 15]. Support stays off this cascade, interpolating a region-uniform institutional floor and a local fiscal-capacity term [30, 22],

$$S_r(\lambda_S) = \text{nz}((1 - \lambda_S)S_0 + \lambda_S(S_0 + \lambda_{\text{tax}}\text{TaxBase}_r)), \quad (6)$$

so the experiments can ask whether support offsets or reinforces hierarchy.

3.3 Real-World Settings

We evaluate five urban-system cases with historical population data sufficient for initialization and forward simulation: China-triad, a multi-core corridor over Shanghai, Beijing, Hong Kong, and Shenzhen–Guangzhou; U.S.-federal, a distributed metropolitan system; UK-devolved, a London-centered system with devolved regional centers; Moscow, a radial capital region; and Tokyo, an integrated mega-region [31]. Each case is initialized from official historical population shares, from which the (O, C, V, S) fields are generated using the functions above. The region unit differs in scale across cases: in China-triad, U.S.-federal, and UK-devolved a region is a metropolitan or sub-national center within a national system, whereas Moscow and Tokyo are treated at a metropolitan scale where a region is a core or peripheral sub-center. The region count R and the unit’s meaning therefore vary across cases, which bears on cross-case MAE comparison.

3.4 Annual Update and Measurement

Our annual update runs perception, choice, mobility, regional feedback, and measurement as one recurrent process, so concentration emerges from coupled agent-region dynamics and not from post hoc clustering. The experiment suite spans feature-function sweeps over opportunity, cost, visibility, support, and connectivity, region-configuration comparisons within one model family, the 2026 hindcast against official shares, and joint reading of concentration with cost and welfare. Cost burden is the population mean of the cost component agents bear in their chosen region relative to a baseline, and aggregate welfare is the population mean

Table 1. Best-fitting model settings initialized from 2010 regional shares, evaluated against official 2026 shares. MAE is the population-share mean absolute error in percentage points, averaged across seeds. Best fit in **bold**.

Region	Opportunity	Cost	Visibility	Support	κ	MAE (pp.)
China-triad	Linear	Linear	Power-law salience	0.50	1.85	1.07
U.S.-federal	Urban scaling	Linear	Power-law salience	0.50	0.80	0.96
UK-devolved	Linear	Linear	Power-law salience	0.50	1.85	1.80
Moscow	Urban scaling	Linear	Power-law salience	0.50	1.85	1.48
Tokyo	Threshold	Linear	Linear salience	1.00	0.80	2.29

realized utility of chosen trajectories; we report both alongside concentration because dispersion can reflect displacement out of an over-expensive core rather than healthier spread. The simulator runs in PyTorch CUDA over batched scenario, seed, and sweep dimensions, with agents in fixed-capacity buffers under active masks, and reproduction runs accept only zero-drop runs.

4 Experiments and Results

We organize the experiments around four claims. First, the model can be calibrated to the 2026 population-share distribution from 2010 data with small error, fitting polycentric and federal structures more closely than single-core ones. Second, the calibrated model projects stronger collapse by 2036, most pronounced in primate and radial systems. Third, both feature functions and regional structure are necessary, since changing either substantially raises mean absolute error (MAE). Fourth, concentration must be read alongside cost burden and welfare, since lower concentration need not imply a welfare gain. All experiments report distributional outcomes over the continuous trajectory-index space.

Table 2. Simulated 2036 outcomes from the 2026 initialization. Higher top share and lower tail mass indicate stronger collapse.

Case	Top share	Tail mass	Eff. neigh.	Top reg. share
China-triad	0.078	0.922	42.83	0.184
U.S.-federal	0.103	0.898	28.71	0.214
UK-devolved	0.181	0.819	14.54	0.330
Moscow	0.560	0.440	4.34	0.763
Tokyo	0.729	0.271	2.88	0.878

4.1 Reproducing 2026 Population Share from 2010 Population Data

For each run, we simulate from 2010 official shares to 2026, average final shares over 10 seeds, and select the suite with the lowest MAE against official 2026 shares. Because selection and comparison use the same target, this is an in-sample calibration, not out-of-sample validation; empirical support comes from differential fit across structures and the transfer ablation in Section 4.3.

Table 1 shows MAE from 0.96 percentage points in U.S.-federal to 2.29 in Tokyo, with China-triad, UK-devolved, and Moscow between 1.07 and 1.80. The

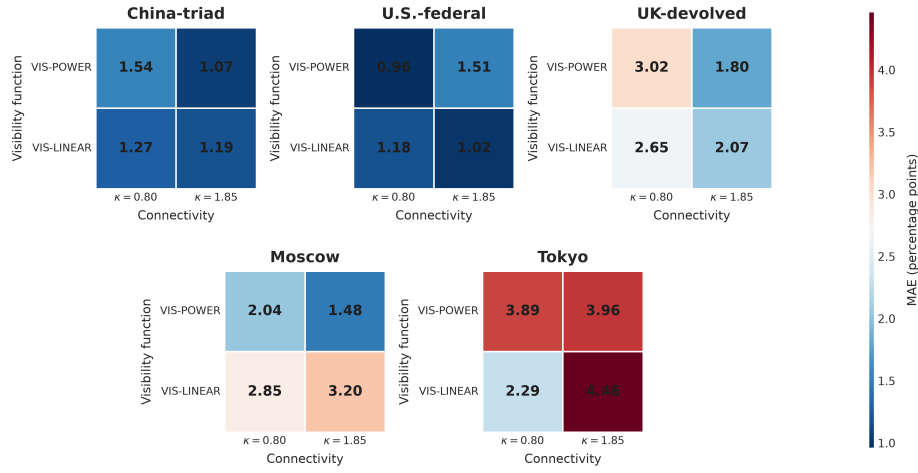


Fig. 2. Connectivity-visibility perturbations by regional case, blue lower MAE and red higher.

best-fitting specifications differ across regions: China-triad and UK-devolved favor linear opportunity, linear cost, power-law visibility, support linkage 0.50, and $\kappa = 1.85$; U.S.-federal and Moscow favor urban-scaling opportunity; and Tokyo fits best under threshold opportunity, linear visibility, full support linkage, and $\kappa = 0.80$. Polycentric and federal cases fit more closely than the most single-core one, indicating that regional structure shapes how the mechanisms behave.

4.2 The Simulation Projects Stronger Trajectory Collapse by 2036

Table 2 reports the 2036 projection from the calibrated models, separating concentration in trajectory space (top trajectory-neighborhood share) from concentration at the regional level (top region share). Concentration is strongest in Moscow and Tokyo, with top trajectory-neighborhood shares of 0.560 and 0.729 against 0.181 in UK-devolved, 0.103 in U.S.-federal, and 0.078 in China-triad; tail mass, effective neighborhoods, and top region share follow the same ordering, with top region shares of 0.878 in Tokyo and 0.763 in Moscow. Conditional on the persistence of current conditions, collapse is most severe in primate and radial systems, while polycentric and federal systems stay substantially more distributed.

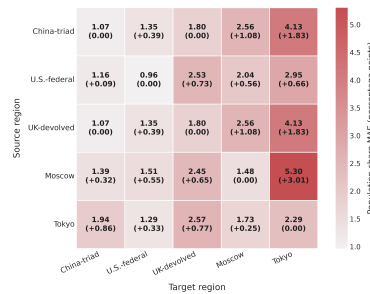


Fig. 3. Cross-region transfer of best-fitting suites, source by row and target by column, darker for larger MAE.

4.3 Ablation Study

Mechanism Design and Regional Structure Are Both Necessary Figure 2 shows that the best connectivity–visibility setting differs by case. China-triad, UK-devolved, and Moscow fit best under high connectivity and high visibility, U.S.-federal under high visibility and low connectivity, and Tokyo under low values of both. This case-specific pattern indicates that connectivity is mediated by visibility and regional structure.

Figure 3 shows the same dependence through cross-case transfer. Specifications transfer well only between China-triad and UK-devolved, which share the same best-fitting suite. Otherwise MAE rises, with the largest loss from applying the Moscow specification to Tokyo. Fit therefore depends on both the mechanism setting and the spatial structure to which it is applied.

Collapse, Cost Burden, and Welfare Must Be Read Together A more dispersed distribution can reflect displacement rather than broader access. Whereas Table 2 gives the headline 2036 projection from the 2026 initialization, Table 3 isolates sensitivity to the initialization year by starting the same Tokyo system from 2010, 2024,

and 2025 shares. The 2010 start yields a lower top share, higher tail mass, and more effective neighborhoods than the 2024 and 2025 starts, yet the highest cost burden and lowest welfare: a 2010 top share of 0.511 against 0.758 and 0.730, with cost burden 1.133 and welfare 1.667, the worst profile of the three. Concentration therefore calls for joint reading with cost burden and welfare, above all when dispersion follows from constrained movement rather than improved opportunity. Because aggregate welfare is mean realized utility on the random-utility scale, which is ordinal and τ -dependent, we read it only within a case across initializations and not as a quantity comparable across cases.

Table 3. Concentration, cost burden, and welfare in Tokyo-to-2036 scenarios.

Init.	Top share	Tail mass	Eff. neigh.	Cost burden	Agg. welfare
2010	0.511	0.489	6.19	1.133	1.667
2024	0.758	0.242	2.66	1.020	1.977
2025	0.730	0.270	2.86	1.011	1.994

5 Discussion

Trajectory concentration depends on the joint configuration of mechanisms and regional structure. The same model family yields distributed outcomes in polycentric and federal systems, but stronger concentration in radial and primate systems. The connectivity–visibility perturbations also differ by case, so concentration cannot be attributed to one mechanism alone. Regional structure matters independently. Cross-case transfer shows that a specification fitting one system often loses fit on another, indicating that hierarchy and topology condition opportunity, visibility, and cost feedback. Concentration must be read with cost burden and welfare. Lower concentration may reflect displacement from an expensive dominant region rather than broader access, as shown by the Tokyo initialization comparison.

Note that the results are stylized with parameters as mechanism settings, with regional units differ in scale, and calibration is in-sample. The model is therefore a scenario framework, not a demographic forecast.

6 Conclusion

We model social collapse as concentration over recurring time-geographic trajectories. The simulations suggest that concentration emerges from the interaction of multiple factors inside the structure. Polycentric and federal systems remain more distributed than radial and primate systems, while concentration and welfare do not necessarily move together. The implication is that preserving diversity requires multiple viable centers of opportunity, support, and activity, rather than connectivity alone.

References

1. Bettencourt, L.M.A., Lobo, J., Helbing, D., Kühnert, C., West, G.B.: Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences* **104**(17), 7301–7306 (2007). <https://doi.org/10.1073/pnas.0610172104>
2. Bisin, A., Verdier, T.: The economics of cultural transmission and the dynamics of preferences. *Journal of Economic Theory* **97**(2), 298–319 (2001)
3. Centola, D.: The spread of behavior in an online social network experiment. *Science* **329**(5996), 1194–1197 (2010). <https://doi.org/10.1126/science.1185231>
4. Chaney, A.J.B., Stewart, B.M., Engelhardt, B.E.: How algorithmic confounding in recommendation systems increases homogeneity and decreases utility. In: *Proceedings of the 12th ACM Conference on Recommender Systems*. pp. 224–232. Association for Computing Machinery, New York, NY, USA (2018). <https://doi.org/10.1145/3240323.3240370>
5. Chorus, C.G., van Wee, B., Geurs, K.T.: Ict, travel behavior and accessibility: A review of literature, a conceptual model and a research agenda. *Journal of Transport and Land Use* **6**(3), 1–16 (2013)
6. Dekker, R., Engbersen, G.: How social media transform migrant networks and facilitate migration. *Global Networks* **14**(4), 401–418 (2014). <https://doi.org/10.1111/glob.12040>
7. Duranton, G., Puga, D.: Micro-foundations of urban agglomeration economies. In: Henderson, J.V., Thisse, J.F. (eds.) *Handbook of Regional and Urban Economics*, vol. 4, pp. 2063–2117. Elsevier (2004)
8. Ellegård, K., Svedin, U.: Torsten hägerstrand’s time-geography as the cradle of the activity approach in transport geography. *Journal of Transport Geography* **23**, 17–25 (2012). <https://doi.org/10.1016/j.jtrangeo.2012.03.023>
9. Epstein, J.M.: *Generative Social Science: Studies in Agent-Based Computational Modeling*. Princeton University Press, Princeton, NJ, USA (2006)
10. Gabaix, X.: Zipf’s law for cities: An explanation. *The Quarterly Journal of Economics* **114**(3), 739–767 (1999). <https://doi.org/10.1162/003355399556133>
11. Galster, G.C., Sharkey, P.: Spatial foundations of inequality: A conceptual model and empirical overview. *RSF: The Russell Sage Foundation Journal of the Social Sciences* **3**(2), 1–33 (2017)

12. Geurs, K.T., van Wee, B.: Accessibility evaluation of land-use and transport strategies: Review and research directions. *Journal of Transport Geography* **12**(2), 127–140 (2004)
13. Glaeser, E.L., Gyourko, J., Saks, R.E.: Urban growth and housing supply. *Journal of Economic Geography* **6**(1), 71–89 (2005)
14. González, M.C., Hidalgo, C.A., Barabási, A.L.: Understanding individual human mobility patterns. *Nature* **453**(7196), 779–782 (2008). <https://doi.org/10.1038/nature06958>
15. Granovetter, M.: Threshold models of collective behavior. *American Journal of Sociology* **83**(6), 1420–1443 (1978)
16. Grimm, V., Berger, U., DeAngelis, D.L., Polhill, J.G., Giske, J., Railsback, S.F.: The odd protocol: A review and first update. *Ecological Modelling* **221**(23), 2760–2768 (2010)
17. de Haas, H.: A theory of migration: The aspirations-capabilities framework. *Comparative Migration Studies* **9**(8) (2021). <https://doi.org/10.1186/s40878-020-00210-4>
18. Hägerstrand, T.: What about people in regional science? *Papers of the Regional Science Association* **24**, 6–21 (1970). <https://doi.org/10.1007/BF01936872>
19. Jefferson, M.: The law of the primate city. *Geographical Review* **29**(2), 226–232 (1939)
20. Krugman, P.: Increasing returns and economic geography. *Journal of Political Economy* **99**(3), 483–499 (1991)
21. Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabasi, A.L., Brewer, D., Christakis, N., Contractor, N., Fowler, J., Gutmann, M., Jebara, T., King, G., Macy, M., Roy, D., Van Alstyne, M.: Computational social science. *Science* **323**(5915), 721–723 (2009). <https://doi.org/10.1126/science.1167742>
22. Ostrom, E.: Polycentric systems for coping with collective action and global environmental change. *Global Environmental Change* **20**(4), 550–557 (2010). <https://doi.org/10.1016/j.gloenvcha.2010.07.004>
23. Redding, S.J., Turner, M.A.: Transportation costs and the spatial organization of economic activity. Working Paper 20235, National Bureau of Economic Research (2014). <https://doi.org/10.3386/w20235>
24. Roback, J.: Wages, rents, and the quality of life. *Journal of Political Economy* **90**(6), 1257–1278 (1982)
25. Robeyns, I.: The capability approach: A theoretical survey. *Journal of Human Development* **6**(1), 93–117 (2005)
26. Salganik, M.J., Dodds, P.S., Watts, D.J.: Experimental study of inequality and unpredictability in an artificial cultural market. *Science* **311**(5762), 854–856 (2006)
27. Schelling, T.C.: Dynamic models of segregation. *Journal of Mathematical Sociology* **1**(2), 143–186 (1971)
28. Sen, A.: *Inequality Reexamined*. Harvard University Press, Cambridge, MA, USA (1992)
29. Song, C., Qu, Z., Blumm, N., Barabási, A.L.: Limits of predictability in human mobility. *Science* **327**(5968), 1018–1021 (2010). <https://doi.org/10.1126/science.1177170>
30. Tiebout, C.M.: A pure theory of local expenditures. *Journal of Political Economy* **64**(5), 416–424 (1956)
31. United Nations Department of Economic and Social Affairs, Population Division: World urbanization prospects 2025. <https://population.un.org/wup/> (2025)